

1. Write the equation of the line that passes through (7,3) and is parallel to $4x + 2y = 10$.

$$\begin{array}{r} -4x \quad -4x \\ \frac{2y}{2} = \frac{10-4x}{2} \end{array}$$

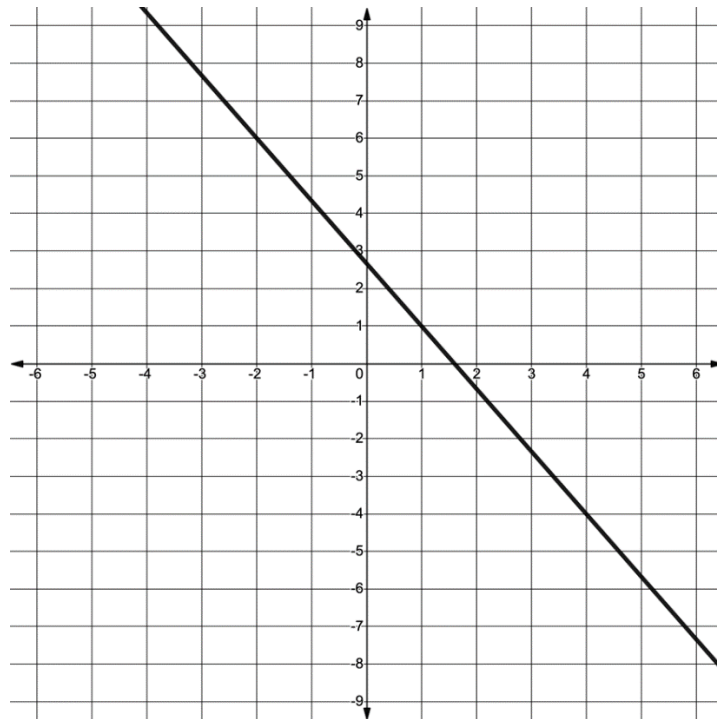
$$y = -2x + 5$$

↑
slope

$$\begin{array}{l} 3 = (-2)(7) + b \\ 3 = -14 + b \\ +14 \quad +14 \\ 17 = b \end{array}$$

$$y = -2x + 17$$

Use the line shown on the coordinate grid to complete questions 2 through 4.



2. What is the slope of the line that is parallel to the line shown on the coordinate grid?

$$-\frac{5}{3}$$

3. Write the equation of the line that is parallel to the line shown on the coordinate grid and that passes through (6, -4).

$$-4 = -\frac{5}{3}(6) + b$$

$$y = -\frac{5}{3}x + 6$$

$$-4 = -10 + b$$

$$+10 \quad +10$$

$$b = 6$$

4. Write the equation of the line that is perpendicular to the line shown on the coordinate grid and that passes through $(6, -4)$.

$$-4 = \frac{3}{5}(6) + b$$

$$-4 = \frac{18}{5} + b$$

$$-\frac{18}{5} - \frac{18}{5}$$

$$b = -7\frac{3}{5}$$

$$y = \frac{3}{5}x - 7\frac{3}{5}$$

5. Write the equation of the line in standard form that is perpendicular to $y = 2x + 5$ and passes through $(-2, 3)$.

slope

$$3 = -2\left(-\frac{1}{2}\right) + b$$

$$3 = 1 + b$$

$$\frac{-1}{2} = b$$

$$y = -\frac{1}{2}x + 2$$

$$+\frac{1}{2}x + \frac{1}{2}x$$

$$2\left(\frac{1}{2}x + y = 2\right)$$

$$x + 2y = 4$$

Use the following information about Line P to complete questions 6 through 8.

Line P passes through $(-23, 33)$ and $\left(\frac{7}{4}, 0\right)$. $\frac{0-33}{\frac{7}{4}-(-23)} = \frac{-33}{\frac{99}{4}} = \frac{-33 \cdot 4}{99} = -\frac{4}{3}$ slope

6. Line W passes through $(0, 9)$ and is parallel to Line P . Write the equation of Line W in slope-intercept form.

$$9 = \frac{-4}{3}(0) + b$$

$$9 = b$$

$$y = -\frac{4}{3}x + 9$$

7. Line T is perpendicular to Line P . What is the slope of Line T ?

$$\frac{3}{4}$$

8. Line T passes through $(-5, 2)$ Write the equation of Line T in all three forms.

Point-Slope Form	Slope-Intercept Form	Standard Form
$y - 2 = \frac{3}{4}(x + 5)$	$2 = -5\left(\frac{3}{4}\right) + b$ $2 = \frac{-15}{4} + b$ $+\frac{15}{4} + \frac{15}{4}$ $\frac{23}{4} = b$ $y = \frac{3}{4}x + \frac{23}{4}$	$\left(y = \frac{3}{4}x + \frac{23}{4}\right) 4$ $4y = 3x + 23$ $\frac{-3x - 3x}{-3x - 3x}$ $-3x + 4y = 23$ or $3x - 4y = -23$

9. Write the equation of the line that is parallel to $9x + y = -2$ and passes through point $(2, -26)$ in slope-intercept form.

$$-26 = 2(-9) + b$$

$$-26 = -18 + b$$

$$+18 \quad +18$$

$$-8 = b$$

$$y = -9x - 8$$

$$-9x \quad -9x$$

$$y = -9x - 2$$

10. Write the equation of the line that is perpendicular to $\underline{x} = \underline{-y}$ and passes through point $(\frac{7}{2}, \frac{7}{2})$.

$$\frac{7}{2} = (+1) \left(\frac{7}{2} \right) + b$$

$$\frac{7}{2} = \frac{7}{2} + b$$

$$-\frac{7}{2} \quad -\frac{7}{2}$$

$$0 = b$$

$$y = x$$

$$-1 \quad -1$$

$$y = -x$$